

Smart City Traffic Optimisation System: A Comprehensive Research Report

Abstract

This capstone report synthesises recent advancements in Intelligent Transportation Systems (ITS) to propose a comprehensive Smart City Traffic Optimisation System. Leveraging real-world data and advanced deep learning models such as Multilevel-Gated Recurrent Units (M-GRU) and Generative AI, this report addresses the critical challenge of urban traffic congestion. By integrating cooperative vehicular rerouting, adaptive traffic signal control, and robust communication frameworks (such as Wi-SUN via Smart Grids), the proposed system aims to reduce travel time, lower fuel consumption, and improve urban mobility. *Note: As requested, system architecture designs involving OpenStreetMap, NotebookLM, ChatGPT, Claude, Gemini, Tableau, Figma, and Notion incorporate external industry-standard knowledge to supplement the uploaded academic sources.*

Executive Summary

Urbanisation has led to severe traffic congestion, necessitating the shift from static traffic management to dynamic, AI-driven Intelligent Transportation Systems (ITS). This project evaluates multiple methodologies, including the maximum weighted flow (MaxPWFlow) adaptive signal method, M-GRU predictive models, and Generative AI simulations. Findings indicate that AI algorithms combined with IoT and edge computing can achieve up to a 30% reduction in congestion. Analysis of the Metro Interstate Traffic Volume dataset reveals that traffic density is heavily influenced by temporal and meteorological factors. A no-code system architecture is proposed, assigning collaborative roles to various Large Language Models (LLMs) to process data, generate predictive insights, and visualise outcomes via Tableau and Figma. The report concludes with future enhancements, focusing on edge AI, federated learning, and ethical considerations regarding data privacy.

1. Core Concepts in Smart City Transportation

- **Intelligent Transportation Systems (ITS):** ITS represents a shift from monolithic, independent management to collaborative system intelligence. It integrates digital

infrastructure, automated transportation tools, and data analytics to facilitate safe and efficient passenger and freight services.

- **Adaptive Traffic Signal Control:** Unlike traditional fixed-time signals, adaptive methods use real-time traffic situations on adjacent lanes to dynamically choose the next traffic light phase. Advanced models include reinforcement learning, the Max Pressure algorithm, and the MaxPWFlow deterministic predictive model which selects phases based on maximum predicted weighted traffic flow.
- **Smart Cities:** Smart cities leverage technologies like the Internet of Things (IoT), artificial intelligence (AI), and data analytics to improve urban living standards. They require robust communication layers, such as Wi-SUN networks, which can share infrastructure with smart grids to support wide-scale smart city applications.
- **Traffic Congestion:** A pervasive problem caused by urban population growth and vehicle increases, resulting in financial losses, time waste, increased fuel consumption, and environmental pollution. Peak congestion typically occurs during morning and evening commuting hours.
- **AI in Transportation:** AI transforms traffic management from reactive measures to proactive, self-learning solutions. Deep learning models like Generative Adversarial Networks (GANs), Transformers, and Reinforcement Learning (RL) are deployed for realistic traffic simulation, predictive analytics, and adaptive signal control. Moreover, Multilevel-Gated Recurrent Units (M-GRU) accurately predict congestion by adapting to the dynamic and nonlinear behaviours of road networks.

2. Comparative Literature Review

Table 1: Comparative Analysis of Uploaded Research

Source Focus	Objectives	Methodology	Dataset Used	AI Techniques	Key Findings	Limitations	Research Gaps
Cooperative Rerouting & Signals	Combine vehicular rerouting with adaptive signal control to	MaxPWFlow signal control + A* algorithm with repulsion (AR*).	SUMO simulation (grid4x4, cologne8).	Deterministic predictive models, heuristic search.	Reduced average waiting time, travel time, and fuel	Relies on simulated environments.	Integration of vehicle trajectory planning algorithms.

	reduce congestion.				consumption.		
GRID-CITY Comm. Framework	Share Smart Grid communication infrastructure with Smart City apps.	Wi-SUN specification, MQTT-SN protocol, IEEE 802.15.4e.	Ipiranga (Brazil) Smart Grid network data.	None explicitly mentioned.	Network operates with <2% duty cycle; viable for hosting smart city IoT.	Potential energy dissipation with frequent communication.	Long data network restart periods following blackouts.
Generative AI Mobility	Explore Generative AI for predicting and optimising traffic flow.	GANs, Reinforcement Learning, Transformers.	GPS, IoT, CCTV, historical data (NYC, Singapore).	GANs, RL, Transformers, Deep Q-Networks.	30% reduction in congestion; 95% accuracy in LA flow prediction.	Computationally intensive, requires massive labelled data.	Integration of edge AI, quantum computing, and federated learning.
M-GRU Traffic Prediction	Predict traffic congestion dynamically.	Multilevel-Gated Recurrent Unit (M-GRU).	Traffic volume, weather conditions, sensor data.	M-GRU, CNN, Bi-LSTM, RNN.	Achieved 0.887 accuracy and 82.34 MAE; outperformed Conv-Bi-LSTM.	Occasional data sparsity leads to inaccuracies.	Needs advanced neural networks with optimisation algorithms.
ITS Standards Architecture	Build an analysis model for urban ITS standard systems.	WSR methodology & Zachman framework.	Current China ITS standard policies.	N/A (Systems architecture).	Designed standard matrix covering life cycle, intelligent units, and types.	Standard promotion and pilot mechanisms are imperfect.	Gaps in key cross-platform integration and big data standards.

Integrated Literature Review

The literature demonstrates a clear evolutionary trajectory in urban mobility from isolated physical solutions to interconnected, data-driven AI systems. Earlier operational optimisations, such as the cooperative application of adaptive traffic signals and rerouting models (A* with repulsion), successfully proved that dynamic, real-time adjustments significantly lower fuel consumption and travel delays. To support the heavy data load required by these dynamic adjustments, resilient communication infrastructure is necessary. Heck et al. demonstrated that existing Smart Grid Wi-SUN infrastructures operate at low duty cycles (<2%) and are economically viable for supporting broader smart city IoT sensors.

Building on this foundational infrastructure, advanced AI models inject predictive capabilities into the system. Generative AI, including GANs and Transformers, is capable of creating highly realistic traffic simulations and anticipating congestion up to two hours in advance with 95% accuracy. Concurrently, Sravani et al. established that time-series traffic prediction is best handled by Multilevel-Gated Recurrent Units (M-GRU), which solve the vanishing gradient problem of traditional RNNs and outperform Bi-LSTM models by adapting dynamically to road networks. Ultimately, the successful deployment of these AI and IoT networks requires a robust standardisation framework; utilizing the WSR methodology and Zachman framework ensures functional compatibility and cybersecurity across the entire intelligent transportation lifecycle.

3. Metro Interstate Traffic Volume Dataset Analysis

Based on the uploaded dataset `Metro_Interstate_Traffic_Volume.csv`, the data records temporal, meteorological, and volume metrics for an interstate highway.

- **Feature Explanation:**

- `traffic_volume`: The dependent target variable representing the number of vehicles passing a point (Traffic Density).
- `holiday`: Categorical indicator of national or regional holidays affecting commuter behaviour.
- `temp`, `rain_1h`, `snow_1h`, `clouds_all`: Numerical weather condition parameters. Weather heavily impacts speed limits and road congestion.
- `weather_main` & `weather_description`: Categorical descriptors (e.g., Clear, Mist, Rain, Snow, Drizzle).
- `date_time`: Timestamps at hourly intervals enabling the extraction of temporal features (rush hour, day of the week).

- **Important Variables:** date_time (crucial for capturing the morning and evening peak hours), weather_main (precipitation directly affects traffic flow), and traffic_volume (the primary metric for congestion scaling).
 - **Possible KPIs:**
 1. Average Hourly Traffic Volume (identifying peak vs. off-peak).
 2. Congestion Risk Index (Traffic volume weighted by adverse weather features like snow_1h).
 3. Traffic Flow Variance (Tracking dynamic nonlinear shifts across days).
 - **Support for Prediction:** This dataset acts as the exact historical training input required by the proposed M-GRU model. By feeding historical weather and vehicle count data into the M-GRU's update and reset gates, the system can evaluate historical dependencies and predict future congestion states (Smooth, Congested, Heavily Congested).
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4. No-Code System Architecture & Workflow

Disclaimer: The implementation specifics of the following tools (Tableau, Figma, Notion, OSM, LLMs) leverage external technical best practices integrated with the uploaded academic concepts to fulfill the capstone requirements.

Architecture Stack:

- **Data Collection & API:** OpenStreetMap (Spatial/Routing data), Metro Interstate Traffic Dataset (Historical/Temporal Data).
- **Knowledge Base & Synthesis:** NotebookLM, Notion.
- **AI Engine (Logic & Prediction):** ChatGPT, Claude, Gemini.
- **Visualisation & UI:** Tableau (Analytics), Figma (Prototyping/UI).

Workflow:

1. **Data Ingestion:** Historical traffic volumes and weather data from the Metro dataset are aggregated alongside geospatial road network graphs from OpenStreetMap.
2. **Data Processing (Gemini & ChatGPT):** ChatGPT generates no-code scripts to clean the data and engineer features (e.g., extracting rush hours). Gemini handles real-time API integrations mapping OSM coordinates to traffic nodes.

3. **Knowledge Retrieval (NotebookLM & Claude):** NotebookLM stores academic literature (M-GRU, MaxPWFlow models). Claude retrieves this context to design the logic rules for the system's traffic routing algorithms.
 4. **Prediction & Decision:** The AI simulates M-GRU predictive logic to classify road segments. If congestion is predicted, an AR* (A* with repulsion) routing logic computes alternative paths to prevent secondary traffic jams.
 5. **Visualisation (Tableau & Figma):** The predicted traffic states and KPIs are exported to Tableau for city manager analytics. Figma designs the end-user interfaces (mobile app for drivers, dashboard for managers).
 6. **Documentation (Notion):** The entire lifecycle, adhering to the Urban ITS Standard Architecture (Basic, Technology, Product, Application categories), is documented in Notion.
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5. Recommended Dashboards, Screens, and Documentation

- **Tableau Dashboards (City Managers):**
 - *Predictive Congestion Map:* A geospatial heat map overlaying OpenStreetMap, highlighting road segments classified as "Heavily Congested" based on M-GRU outputs.
 - *Weather Impact Matrix:* Scatter plots correlating rain_1h/snow_1h from the Metro dataset with traffic_volume drops.
 - *System Duty Cycle Monitor:* IoT health tracking based on the GRID-CITY framework to monitor Wi-SUN network load (<2% duty cycle KPI).
- **Figma Screens (End Users/Drivers):**
 - *Real-Time Navigation Screen:* Displays the current route, predicted travel time, and dynamic rerouting prompts generated by the A* repulsion algorithm.
 - *Urgency Ranking Notification:* Alerts drivers of alternative routes based on their "urgency metric" (remaining travel time vs destination distance).
- **Notion Documentation Structure:**
 - Basic Universal: Terminology and data dictionaries (Metro dataset schema).
 - Digital Infrastructure: Details on Wi-SUN and MQTT-SN protocols.

- Big Data Management: M-GRU model weights, hyperparameters, and OpenStreetMap API endpoints.
 - Integrated Transportation Management: Logic documentation for the MaxPWFlow adaptive signal control algorithms.
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6. AI Traffic Optimisation Engine Collaboration

To optimise traffic efficiently, the LLMs act as a collaborative, agentic ensemble:

- **Claude (The Architect & Strategist):** Due to its large context window and strong reasoning, Claude ingests academic research (ITS standards, M-GRU mathematics) and designs the overarching system rules. It dictates *when* adaptive signals should override fixed signals based on the literature.
 - **ChatGPT (The Data & Logic Engineer):** Acts as the execution engine. ChatGPT writes the underlying logic for the AR* (A* with repulsion) algorithm and calculates the "weighted footprint" of vehicles to prevent rerouted traffic from causing new bottlenecks. It also structures the Metro CSV data for the dashboards.
 - **Gemini (The Real-Time Multimodal Analyst):** Leverages its capability to process live web data. Gemini integrates real-time weather APIs and OpenStreetMap disruptions, comparing live conditions against the Metro historical dataset to feed real-time inputs into the predictive engine.
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7. Project Limitations, Enhancements, and Ethics

- **Limitations:**
 - *Computational Complexity:* Generative AI models (GANs) and M-GRU networks are computationally expensive and require vast amounts of labelled data, which can strain smart city edge devices.
 - *Data Sparsity:* Dynamic and nonlinear road network changes, alongside map matching errors or zero speed adjustments, can lead to inaccurate predictions.
- **Future Enhancements:**
 - Integration of Quantum computing and Federated Learning to enhance traffic intelligence without centralising sensitive data.

- Deployment of Edge AI on Roadside Units (RSUs) to allow immediate, localised traffic signal adjustments without cloud latency.
 - **Ethical Considerations:**
 - Data privacy concerns arising from continuous tracking of vehicle GPS and CCTV camera imagery.
 - Algorithmic biases in AI-driven public transport scheduling that might disenfranchise certain urban demographics.
 - **Expected Outcomes:**
 - Attainment of high prediction accuracy (targeting M-GRU's ~0.887 benchmark).
 - Reduction in average travel times, waiting times, and greenhouse gas emissions due to optimized fuel consumption.
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8. Capstone Deliverables

Abstract (Short)

This capstone project conceptualises a Smart City Traffic Optimisation System utilizing Multilevel-Gated Recurrent Units (M-GRU) and adaptive signal control (MaxPWFlow). By integrating historical weather and traffic volume data, the system predicts congestion and executes cooperative vehicular rerouting. The project outlines a no-code LLM-driven architecture supported by Wi-SUN IoT frameworks, aiming to deliver safer, more sustainable urban mobility while mitigating computational and ethical challenges.

Conclusion

Generative AI and advanced deep learning represent a paradigm shift in urban traffic management. By transitioning from static infrastructure to dynamic, predictive ecosystems driven by M-GRU networks and adaptive routing, smart cities can significantly alleviate congestion. Implementing this capstone's architecture—backed by robust ITS standards and IoT communication—promises measurable benefits in travel efficiency and environmental sustainability, provided data privacy and computational barriers are carefully managed.

15 Viva Questions & Answers

Q1: What is the primary advantage of the M-GRU model over traditional RNNs or Bi-LSTMs in traffic prediction? *Answer:* M-GRU effectively handles time-series data and overcomes the vanishing gradient problem of RNNs using reset and update gates. It uses multilevel layers to

adapt to the dynamic changes in road networks, achieving higher accuracy (0.887) and lower errors than Conv-Bi-LSTM models.

Q2: How does the MaxPWFlow algorithm improve upon standard traffic lights? *Answer:*

MaxPWFlow is an adaptive control method that selects the next traffic light phase based on predicting the maximum weighted traffic flow of vehicles approaching the intersection, rather than using fixed time intervals.

Q3: Why is the Wi-SUN specification recommended for Smart City traffic infrastructure?

Answer: Wi-SUN provides a robust, wireless Field Area Network suitable for Neighborhood Area Networks. It operates efficiently out of doors with very low duty cycles (under 2%), leaving ample network capacity to host smart city IoT sensors alongside smart grid meters.

Q4: In vehicular rerouting, what is the purpose of the "urgency metric"? *Answer:* Urgency is the difference between a vehicle's remaining travel time in current congestion versus travel time at maximum allowed speed. Vehicles closer to their destinations or facing worse delays rank higher in urgency and are rerouted first.

*with repulsion (AR)" algorithm?** *Answer:* It is a modified pathfinding algorithm that considers the planned routes of all other vehicles in the network as a "repulsive force." This prevents rerouting systems from accidentally sending multiple vehicles to the same alternate route, creating a new traffic jam.

Q6: According to the literature, how does weather impact traffic congestion prediction?

Answer: Weather conditions strictly impact vehicle speed limits and overall road capacity. Feeding meteorological data (like rain or snow) into prediction models drastically improves the accuracy of congestion forecasting.

Q7: What is the Zachman framework, and how is it used in this project? *Answer:* The Zachman framework is an enterprise architecture model used to describe systems from multiple perspectives. It is used to structure the Urban ITS Standard System, aligning transport domains, life cycles, intelligent units, and standard types logically.

Q8: What specific challenges do Generative AI models face when deployed for traffic management? *Answer:* They are computationally intensive, require massive amounts of labelled training data, and present significant data privacy and ethical concerns when processing real-time citizen movements.

Q9: How is 'Traffic Density' defined in the context of the M-GRU prediction model? *Answer:* It is determined by estimating the number of vehicles on a road segment based on vehicle parking length or loop detector occupancy. The relationship between speed, flow, and density is the core formula used to feed the M-GRU model.

Q10: What role does Edge Computing play in next-generation traffic solutions? *Answer:* Edge AI models deployed directly on Roadside Units (RSUs) or traffic cameras process vast amounts of data locally and in real-time, reducing cloud latency and accelerating adaptive signal adjustments.

Q11: Why does the proposed system framework split ITS standards into four specific 'Intelligent Units'? *Answer:* To comprehensively categorise standardisation targets. The units are Resources, Equipment, Data, and Platforms, ensuring all layers from physical sensors to cloud analytics are governed by interoperable protocols.

Q12: How did the integration of AI in Singapore impact their traffic systems? *Answer:* Singapore implemented deep reinforcement learning for dynamic traffic light adjustments, achieving a 30% reduction in congestion.

Q13: What protocol acts as the adaptation layer for the smart city IoT devices connecting to the Smart Grid? *Answer:* MQTT-SN (Message Queuing Telemetry Transport for Sensor Networks) is used. It is designed for resource-constrained environments, operating over UDP with smaller messages, adapting perfectly to the Wi-SUN backhaul.

Q14: Explain the difference between 'Fast Research' and 'Deep Research' regarding future traffic pattern discovery. *Answer:* While not directly in the text, applying general AI data discovery concepts, fast research provides immediate heuristic approximations of traffic patterns, whereas deep research requires extensive computational cycles (like GAN simulations) to identify complex, nonlinear traffic dependencies over several minutes.

Q15: What is the significance of integrating public transport scheduling with AI, as seen in Tokyo? *Answer:* Generative AI optimised metro scheduling and developed a real-time passenger forecasting system, which increased overall public transport efficiency by 25%